

Markscheme

Mathematics: analysis and approaches
Practice question bank

47. (a)

Given that $\log_a 2 = 0.3562$ and $\log_a 3 = 0.5646$, both correct to 4 decimal places.

Find the value of a .

rewrite in exponential form: $a^{0.3562} = 2$ (M1)

$$a = 2^{\frac{1}{0.3562}} \quad (M1)$$

$$a = 6.99 \dots \quad \text{A1}$$

$$a = 7 \quad \text{A1}$$

[4 marks]

(b)

Hence find the value of $\log_a 6$.

$$\log_a 6 = \log_a 2 + \log_a 3 \quad (M1)$$

$$= 0.3562 + 0.5646 = 0.9208 \quad \text{A1}$$

[2 marks]